The connections between the Intelligence Science and 《Tenet》

—Causality, Temporal Cognition, and the Nature of Intelligence

What if the future were not something we move towards, but something that already exists?

I have watched Christopher Nolan's *Tenet* eight times. Yet, every time I revisit the film, I find myself confronting questions that seem increasingly difficult to answer. At first glance, *Tenet* appears to be a complex science-fiction thriller built around temporal inversion. But beneath its intricate narrative lies something more fundamental: a philosophical exploration of time, causality, knowledge, and human agency.

This raises a question： **Can intelligence truly understand time, or can it only learn to predict what happens within it?**

I believe this question offers an interesting perspective from which to explore the foundations of Intelligence Science.

1. The Arrow of Time: Is Time Fundamentally Asymmetric?

One of the central concepts in *Tenet* is inversion. Objects and people can move in a direction opposite to the ordinary temporal progression of the surrounding world.

Although this mechanism is fictional, it touches upon a genuine problem in physics: the arrow of time.

Many fundamental physical equations exhibit time-reversal symmetry. In simplified terms, certain physical processes remain mathematically valid when the direction of time is reversed. However, our macroscopic experience of the world is strongly asymmetric.

We remember the past, not the future. We observe broken glasses, but rarely see them spontaneously reconstruct themselves. Heat flows naturally from hotter objects to colder ones.

The second law of thermodynamics provides a statistical explanation for this asymmetry:

$$\Delta S_{\text{total}} \geq 0.$$

Entropy tends to increase in an isolated system.

This distinction is important. Time-reversal symmetry in physical laws does not imply that macroscopic time can simply be reversed. Nor does it establish the physical possibility of the inversion mechanism portrayed in *Tenet*.

Nevertheless, the film invites us to consider an interesting thought experiment:

If the direction of physical processes could be reversed, how would an intelligent agent perceive and understand its environment?

Would its memories remain consistent? Would its predictions still be meaningful? And how would it construct a model of a world whose temporal structure no longer matches its own experience?

These questions lead us from physics towards cognition.

2. Causality: Understanding Why, Not Merely What

Perhaps the most interesting dimension of *Tenet* is not time itself, but causality.

In our ordinary understanding of the world, causes precede effects. We construct explanations by identifying relationships between events and determining how one event influences another.

However, *Tenet* deliberately challenges this intuitive ordering.

The film introduces situations in which an effect appears to precede its cause from the perspective of an ordinary observer. This creates a distinction between temporal ordering and causal structure.

It is important not to confuse the two. A reversed temporal experience does not automatically establish that causation itself has been reversed.

This distinction connects to a fundamental problem in artificial intelligence and machine learning.

A system may identify statistical correlations and produce highly accurate predictions without understanding the underlying causal mechanisms.

For example, observing that two variables frequently change together does not establish that one causes the other.

Judea Pearl's framework of causal inference provides a mathematical language for distinguishing observation from intervention.

In particular, the intervention operator:

$$P(Y|do(X = x))$$

represents the distribution of an outcome (Y) when we actively intervene to set (X=x), rather than merely observe (X=x).

This distinction is fundamental to causal reasoning.

An intelligent system should ideally do more than recognize patterns. It should develop an understanding of how changes in one part of the world influence other parts.

In this sense, *Tenet* offers an unusual thought experiment: what happens when an agent encounters events whose causal relationships cannot be understood through ordinary temporal intuition?

Perhaps intelligence requires more than learning the sequence of events. Perhaps it requires reconstructing the mechanisms that generate them.

3. World Models and Temporal Cognition

An intelligent agent does not interact with the world through isolated observations alone.

It must maintain some representation of its environment, infer hidden states, anticipate possible outcomes, and select actions accordingly.

This is closely related to the concept of a world model.

A simplified representation of an intelligent agent's decision-making process can be written as:

$$s_t \rightarrow a_t \rightarrow s_{t+1},$$

where:(s_t) represents the current state;

(a_t)represents the action selected by the agent;

(s_{t+1})represents the subsequent state.

The agent learns from the relationship between actions and environmental transitions.

In reinforcement learning, this process is often formalized through a Markov Decision Process.

However, the deeper question is not simply whether an agent can predict the next state.It is whether the agent can construct a sufficiently accurate internal model of the world to reason about hypothetical situations.

What would happen if a different action were selected? What would happen if an external condition changed? Which outcomes are possible, and which are merely imagined?

These questions connect temporal cognition with planning, counterfactual reasoning, and model-based decision-making.

In *Tenet*, characters must reason about events from different temporal perspectives. Their understanding of the situation depends not only on what they observe, but also on how they interpret the temporal and causal structure of those observations.

This is a fictional scenario, but it highlights a genuine challenge for intelligent systems:

To navigate the future, an agent must construct a model of how the world evolves.

4. Determinism, Free Will, and Intelligent Agency

The philosophical dimension of *Tenet* becomes particularly interesting when considering the statement:

What's happened, happened.

This phrase captures much of the film's underlying philosophy.

If events are already part of a consistent causal structure, can an individual genuinely change the future?

And if an agent's actions are already included in that structure, does it still possess free will?

These questions have occupied philosophers for centuries.

Determinism suggests that events are determined by prior conditions and the governing laws of nature. However, determinism should not automatically be equated with fatalism.

The distinction matters.

Fatalism implies that an outcome will occur regardless of what an individual does. Determinism, by contrast, allows an individual's decisions to be part of the causal process that produces that outcome.

An agent can therefore make meaningful decisions even in a deterministic world.

From the perspective of Intelligence Science, this raises another question:

What makes an intelligent agent autonomous?

Is autonomy simply the ability to select an action? Or does it require the capacity to understand the environment, evaluate alternatives, anticipate consequences, and act according to internally represented objectives?

I would argue that predictive ability alone is insufficient.

An agent that can accurately predict an outcome is not necessarily capable of controlling it. Likewise, an agent that can select actions is not necessarily capable of understanding the causal consequences of those actions.

This leads to an important conceptual distinction:

$$Prediction \neq Understanding \neq Control.$$

These capabilities may be related, but they are not interchangeable.

5. Intelligence as the Understanding of Time and Causality

Returning to the original question, what does *Tenet* have to do with intelligence?

I believe the connection lies in the relationship between intelligence, time, and causality.

An intelligent agent must operate within a changing environment. It must learn from past experiences, interpret present observations, anticipate future possibilities, and make decisions under uncertainty.

These processes are fundamentally temporal.At the same time, meaningful reasoning requires more than temporal prediction. It requires an understanding of the mechanisms through which events influence one another.

This suggests a possible research perspective:

Perhaps one important dimension of intelligence is the ability to construct and reason with causal models of a dynamic world.

Such a perspective connects several fields:

Physics, through the structure and direction of time;

Mathematics, through probability and dynamical systems;

Philosophy, through knowledge, agency, and free will.

Of course, this is not a complete definition of intelligence. Intelligence is a much broader phenomenon, and its foundations cannot be reduced to temporal cognition or causal reasoning alone.

However, these connections deserve serious investigation.

Conclusion: Beyond Prediction

After watching *Tenet* eight times, I still cannot claim to have fully understood every detail of its narrative.

But perhaps understanding the film was never the most interesting objective.

What fascinates me is the set of questions it leaves behind.

We often associate intelligence with the ability to solve problems, recognize patterns, and predict future events. Yet, perhaps a deeper form of intelligence involves understanding why the world behaves as it does, how its underlying mechanisms operate, and how an agent can act meaningfully within those mechanisms.

Prediction tells us what may happen.

Causal reasoning helps us understand why it happens.

Decision-making allows us to participate in what happens next.

And perhaps the pursuit of Intelligence Science begins with understanding the relationship between all three.*What's happened, happened.*

But understanding why it happened, and what it means for what happens next, remains an open intellectual challenge.